SOFTWARE ENGINEERING

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem: Statement #35

CUSTOMIZABLE SUBSCRIPTION BOX SCHEDULER

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1. **Problem Statement**

Customizable Subscription Box Scheduler

A subscription box portal where subscribers customize monthly product selections such as books, coffee, and snacks based on preference tags, with support for pausing or skipping billing cycles.

Target Actors/Stakeholders:

1. Subscriber
2. Fulfillment Lead
3. Payment Gateway

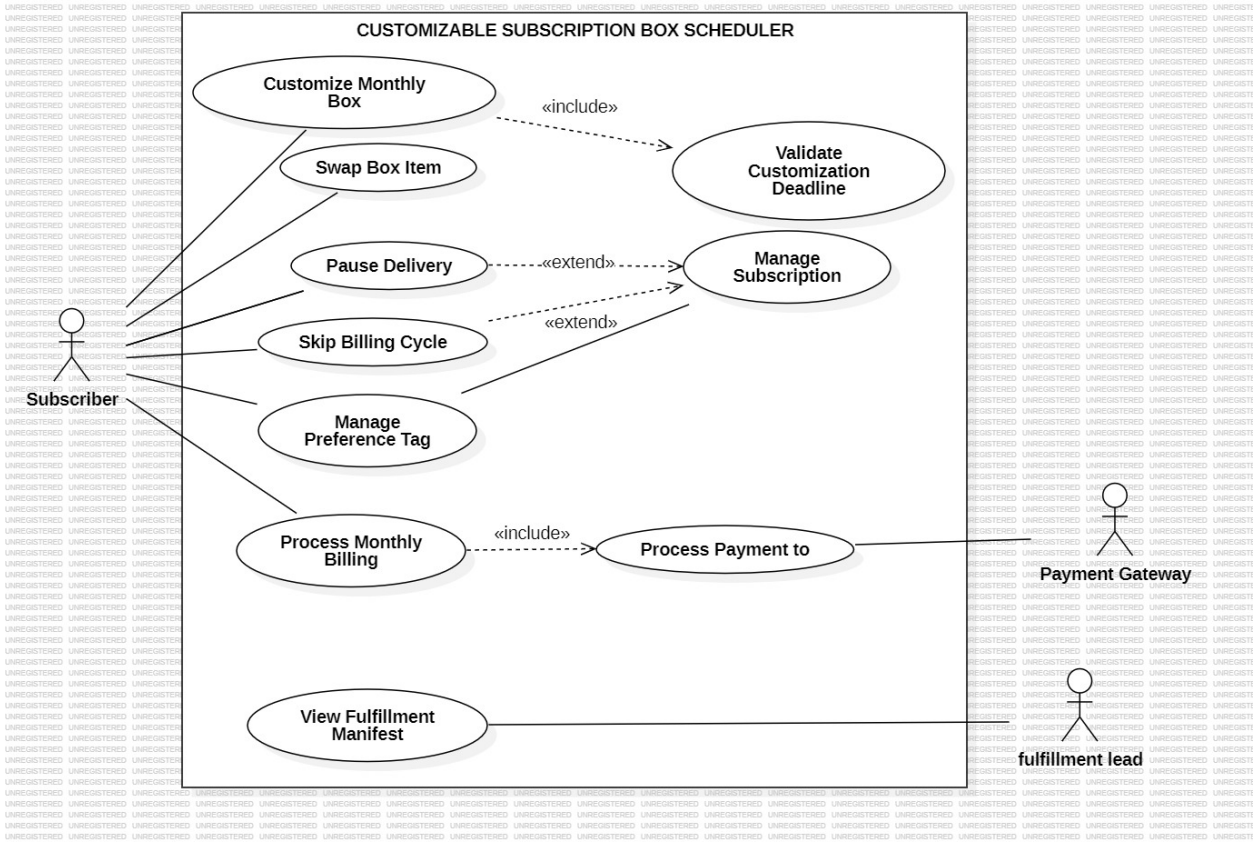
1**.Requirements Table**

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow subscribers to swap monthly box items or pause deliveries up to 48 hours prior to the monthly billing renewal date.	High	**Pass:** The subscriber can swap an item or pause delivery before the 48-hour deadline and the change is saved. **Fail:** The system allows the change after the deadline or does not save it.	Provides the core subscription customization functionality.	Given requirement from problem statement.
FR-002	Functional	The system shall allow subscribers	High	**Pass:** Selecting a preference tag	Enables personalized	Based on product

		to select monthly products based on available preference tags such as books, coffee, and snacks.		displays matching products for selection. **Fail:** Matching products are not displayed.	subscription boxes.	preference functionality.
FR-003	Functional	The system shall allow subscribers to skip a selected monthly billing cycle before the applicable renewal deadline.	High	**Pass:** The selected billing cycle is marked as skipped and no box is scheduled for that cycle. **Fail:** The cycle remains active after a valid skip request.	Supports the skip billing-cycle functionality.	Deadline must be checked.
FR-004	Functional	The system shall update the fulfillment manifest whenever a subscriber successfully changes the contents of a monthly box.	High	**Pass:** The updated product selection appears in the fulfillment manifest after the change. **Fail:** The old selection remains in the manifest.	Ensures fulfillment uses the latest box selection.	Manifest should reflect successful changes.
FR-005	Functional	The system shall allow the fulfillment lead to view the monthly fulfillment manifest containing the current subscriber box selections.	Medium	**Pass:** The fulfillment lead can view the current selections for active subscriptions. **Fail:** The manifest cannot be accessed or shows outdated selections.	Supports accurate preparation of subscription boxes.	Access should be authorized.
NFR-001	Nonfunctional	The fulfillment manifest generator shall export shipping labels for 10,000 monthly boxes in under 1 minute.	High	**Pass:** Shipping labels for 10,000 boxes are exported in less than 60 seconds. **Fail:** Export takes 60 seconds or longer.	Ensures acceptable performance during large-scale fulfillment.	Given requirement from problem statement.
NFR-002	Nonfunctional	The system shall ensure that only authenticated subscribers can	High	**Pass:** An authenticated subscriber can modify only their	Protects subscriber data and prevents unauthorized	Security requirement.

		modify their own subscription selections and billing-cycle preferences.		own subscription. **Fail:** An unauthenticated user or another subscriber can modify it.	changes.	

2.**UML Use-Case Diagram**



A. **Actors**

- Subscriber
- Fulfillment Lead
- Payment Gateway

B. **Use Cases**

- Customize Monthly Box
- Swap Box Item
- Pause Delivery
- Skip Billing Cycle
- Manage Preference Tag
- Manage Subscription
- Validate Customization Deadline
- Process Monthly Billing
- Process Payment
- View Fulfillment Manifest

3.**Use-Case Flow**

Main Success Scenario

- The subscriber selects Customize Monthly Box.
- The system displays the current monthly box items and available products.
- The subscriber selects preference tags such recharge ,bills ,loan's .
- The system displays products matching the selected preference tags.
- The subscriber selects or swaps the desired products.
- The system validates that the customization request is made at least 48 hours before the monthly billing renewal date.
- The system saves the updated monthly box selection.
- The system updates the fulfillment information with the new box selection.
- The system displays a confirmation message to the subscriber.
- The use case ends successfully.

Customization Deadline Passed

- The system determines that the request is within 48 hours of the monthly billing renewal date.
- The system rejects the customization request.
- The system displays a message indicating that customization is no longer available for the current billing cycle.
- The existing monthly box selection remains unchanged.
- The use case ends.